A Heterogeneous UAV UGV Methodology for Aerial Tracking of Mini Autonomous Cars

Santiago Fernandez
Robotics and AI Lab
Technological University of Uruguay
Rivera, Uruguay
santiago.fernandez@estudiantes.utec.edu.uy

Angel da Silva
Robotics and AI Lab
Technological University of Uruguay
Rivera, Uruguay
angel.dasilva@estudiantes.utec.edu.uy

Hiago Sodre
Robotics and AI Lab
Technological University of Uruguay
Rivera, Uruguay
hiago.sodre@utec.edu.uy

Pablo Moraes
Robotics and AI Lab
Technological University of Uruguay
Rivera, Uruguay
pablo.moraes@utec.edu.uy

Bruna Guterres
Robotics and AI Lab
Technological University of Uruguay
Rivera, Uruguay
bruna.devargas@utec.edu.uy

Ricardo B. Grando
Robotics and AI Lab
Technological University of Uruguay
Rivera, Uruguay
ricardo.bedin@utec.edu.uy

***Abstract*—Aerial tracking of autonomous ground vehicles can provide situational awareness when fixed external perception infrastructure is unavailable, but visual-only UAV tracking is limited by field of view, occlusions, and temporary target loss. This paper presents a heterogeneous UAV–UGV methodology for simulated aerial tracking in ROS 2 and Gazebo Sim using a Tello-like UAV and LIMO-like UGVs. Three approaches are evaluated: a color baseline, a YOLO visual baseline, and a UAV–UGV fusion method that combines YOLO target estimates with shared UGV state information during visual gaps. Four scenarios are considered: single-car tracking, crossing trajectories, occlusion, and UAV-assisted awareness. Across 600 simulation runs, the fusion method achieves nearly continuous track coverage and reduces RMSE relative to YOLO by 37.3%, 64.2%, 62.6%, and 61.8% across the four scenarios. Conflict-awareness lead times of 10.20 s and 9.07 s are obtained in S2 and S4, respectively. Under the evaluated simulation conditions, heterogeneous UAV–UGV state sharing improves tracking continuity under intermittent visual observations.**

***Index Terms*—Heterogeneous robotics, UAV-UGV cooperation, aerial tracking, autonomous ground vehicles, object detection, YOLO, sensor fusion.**

I. Introduction

Heterogeneous robotic teams combine platforms with complementary sensing, mobility, and computational capabilities. In air–ground systems, unmanned aerial vehicles (UAVs) provide an elevated viewpoint and wide-area perception, while unmanned ground vehicles (UGVs) offer stable ground-level motion, longer operational persistence, and direct interaction with the environment. This complementarity has motivated increasing interest in UAV–UGV cooperation for inspection, mapping, monitoring, search-and-rescue, and collaborative autonomy [1]–[3].

Aerial tracking of autonomous ground vehicles can be particularly useful in environments where fixed external perception infrastructure is unavailable. This aerial perspective can support monitoring of multiple vehicles, detection of potentially hazardous interactions, fleet monitoring, collision prevention, large-area inspection, and assistance to ground vehicles whose perception is limited by obstacles, short-range sensors, or unfavorable environmental conditions.

Despite these advantages, purely visual tracking is limited by the camera field of view, viewpoint changes, partial occlusions, and target appearance variations. These limitations become more critical in multi-robot scenarios where ground vehicles may cross paths, become partially occluded, or enter shared areas where early conflict awareness is required.

Real-time object detection has become a standard component of robotic perception systems. The YOLO (You Only Look Once) family is widely adopted because it combines accurate object localization with real-time inference [4]. Recent implementations such as YOLOv8 provide practical training and deployment pipelines for robotic applications [5]. However, continuous target awareness may not be maintained through visual detection alone. Visual observations become unavailable when the tracked UGV leaves the camera field of view or becomes temporarily occluded. Under these conditions, a UAV relying only on visual perception may lose target-state availability even when the detector provides accurate observations while the vehicle is visible.

This paper proposes a heterogeneous UAV–UGV methodology for simulated aerial tracking of small-scale autonomous ground vehicles. The system employs a mini UAV with a front-facing RGB camera to observe UGVs in ROS 2 and Gazebo Sim. Three tracking approaches are evaluated: a training-free color baseline, a visual baseline based on YOLO, and the proposed heterogeneous UAV–UGV approach that integrates visual perception with information shared by the ground vehicle. In this work, heterogeneous cooperation refers specifically to inter-platform state sharing for tracking support; joint motion planning and coordinated control are outside the scope of the study. Figure 1 summarizes the proposed heterogeneous UAV–UGV tracking methodology. The simulated environment provides two complementary information streams. The first is

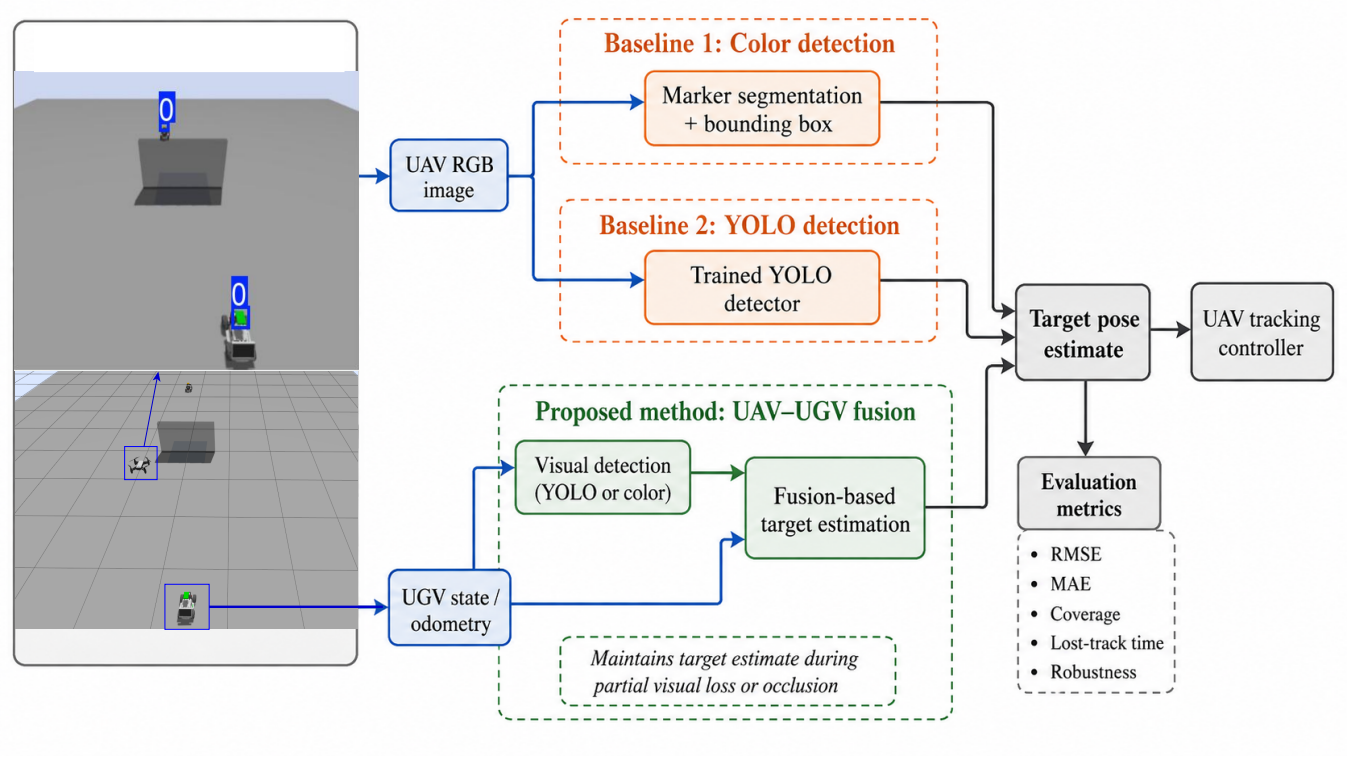


Fig. 1: Overview of the proposed heterogeneous UAV–UGV aerial tracking methodology. The UAV provides front RGB images for visual detection, while UGV state information is used by the proposed fusion method to maintain target estimates during visual gaps. The same simulation environment is used to evaluate the color baseline, the YOLO baseline, and the proposed UAV–UGV fusion approach. Code availability: https://github.com/ricardoGrando/limo_tello_jazzy_spawn. Video: https://www.youtube.com/watch?v=f7xAwTxisKk

the UAV front RGB image. The second is the UGV state information. The RGB stream is processed by both visual baselines. The proposed method combines visual detections with UGV state information to estimate the target position when visual observations become unavailable. This design isolates the effect of shared UGV state information on tracking continuity.

The main contributions of this work are:

- A ROS 2/Gazebo Sim methodology for heterogeneous UAV–UGV aerial tracking of small-scale autonomous ground vehicles.
- A comparison between a color-based visual baseline, a YOLO visual baseline, and a UAV–UGV fusion approach.
- An evaluation across four scenarios: single-car tracking, crossing trajectories, occlusion, and UAV-assisted awareness.
- Experimental evidence that the proposed fusion method increases track coverage to nearly 100% while reducing localization error across all evaluated scenarios.

II. Related Work

A. UAV Target Tracking

UAV target tracking has been studied as both an active and passive perception problem. In active tracking, the UAV adjusts its motion to maintain the target within its field of view, while passive tracking focuses on estimating information about the target from the available observations. State-estimation and fusion strategies, such as Kalman filtering, particle filtering, and learning-based tracking, are commonly used to improve robustness under noisy or intermittent observations [6].

Vision-based UAV tracking systems often combine object detection with temporal filtering. Barisic et al. presented a vision-based system for real-time detection and following of UAVs using YOLO and a Kalman filter, showing that combining visual detections with dynamic estimation improves tracking stability [7]. These works support the idea that visual detections can benefit from complementary state information when observations are intermittent.

B. YOLO-Based Robotic Perception

YOLO is a family of single-stage detectors capable of estimating object locations and classes within a single inference step [4]. Its balance between detection accuracy and processing speed makes it particularly suitable for robotic applications requiring real-time visual perception. YOLOv8 is distributed through the Ultralytics framework, which provides pretrained models, training utilities, and deployment tools [5]. Recent work has analyzed YOLOv8 and reported its practical

advantages for real-time object detection, including its anchor-free design and efficient training pipeline [8].

In UAV applications, YOLO-based detectors have been used for flying-object detection, small-object recognition, and lightweight onboard perception. Reis et al. evaluated YOLOv8 for real-time flying-object detection and emphasized challenges related to small target size, motion, occlusion, and cluttered backgrounds [9]. These challenges are also present in UAV–UGV tracking, where temporary occlusions or field-of-view limitations may interrupt visual observations of the target.

C. Heterogeneous UAV–UGV Cooperation

Heterogeneous UAV–UGV systems have been studied for cooperative localization, mapping, planning, and semantic collaboration. Recent surveys describe air–ground heterogeneous systems as a robotic paradigm in which UAVs and UGVs provide complementary capabilities in perception, mobility, and mission execution [1]. Other works have explored air–ground cooperation for semantic mapping and navigation, where aerial perception provides information that improves ground-robot decision-making [2].

However, it remains important to evaluate under controlled conditions to what extent shared UGV state information can preserve aerial tracking continuity during temporary visual loss.

The present work follows this research direction but focuses specifically on aerial tracking of small-scale autonomous ground vehicles. Unlike a purely visual method, the proposed approach uses the same visual detector as the YOLO baseline and combines visual detections with shared UGV state information to maintain target-state availability during temporary visual loss. This design isolates the contribution of shared UGV state information from that of the learned visual detector.

III. Methodology

The proposed methodology (Figure 1) is implemented in ROS 2 Jazzy and Gazebo Sim using a Tello-like UAV with a front-facing RGB camera and one or more AgileX LIMO-like UGVs. The experimental pipeline comprises a scenario controller, visual detector, tracking module, and experiment logger/evaluator. Three approaches are evaluated: a color baseline, a YOLO visual baseline, and the proposed UAV–UGV fusion method.

A. Color Baseline

The color baseline is a training-free visual method. It detects a colored marker placed on the top of the LIMO using RGB thresholding and connected-component filtering. This method is included as a classical baseline because it does not require a learned detector or a training dataset. However, it is expected to be sensitive to viewpoint changes, marker visibility, lighting, and geometric projection errors.

B. YOLO Visual Baseline

The YOLO baseline uses a trained YOLOv8 detector to identify the LIMO marker from the UAV RGB image.

Bounding-box labels were generated directly from the rendered RGB images by segmenting the visible colored marker attached to each LIMO. The final training labels were therefore image-based rather than obtained through geometric projection from robot state.

Training images were collected from S1–S3 using five fixed random seeds, with S4 reserved for detector validation; dataset collection and tracking evaluation were performed in separate runs.

During evaluation, detections are converted into planar target positions using a front-camera apparent-size range model based on the detected bounding-box width and image bearing. The calibrated model uses an effective marker width of 0.18 m and a forward range bias of 1.09 m.

The YOLO baseline uses only visual information and does not use shared UGV state information when detections are unavailable. In multi-UGV scenarios, visual detections are associated with the nearest available UGV state in the ground plane. Thus, the current implementation evaluates target-state continuity rather than appearance-based multi-object re-identification.

C. Proposed UAV–UGV Fusion Method

The proposed method uses the same trained YOLO detector as the YOLO baseline and incorporates shared UGV state information. When both sources are available, the target position is estimated through complementary weighted fusion:

$$\hat{\mathbf{p}}_k = \alpha \mathbf{p}_k^{UAV} + (1-\alpha)\mathbf{p}_k^{UGV}, \tag{1}$$

where $\mathbf{p}_k^{UAV}$ is the position estimated from the UAV image, $\mathbf{p}_k^{UGV}$ is the idealized simulated UGV state, and a fixed $\alpha = 0.65$ is used in all experiments.

A visual detection is considered valid for up to 0.5 s after reception. When no valid visual observation is available, the tracker directly uses the current UGV state,

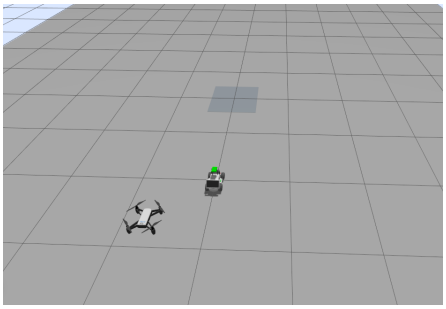
$$\hat{\mathbf{p}}_k = \mathbf{p}_k^{UGV}. \tag{2}$$

If the UGV state is unavailable but a valid visual detection exists, the visual estimate is used directly. Therefore, the method preserves target-state availability during visual gaps without applying a dynamic prediction model. Since the same YOLO detector is used in both approaches, differences relative to the YOLO baseline isolate the contribution of heterogeneous state sharing.

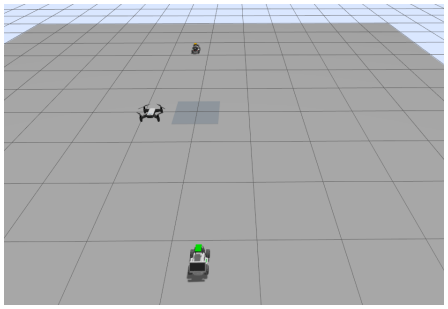
D. Experimental Scenarios

Four scenarios are evaluated, as summarized in Table I. Each method is evaluated over 200 runs: four scenarios, five random seeds, and ten trials per seed.

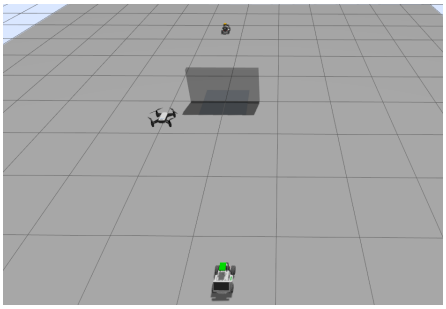
Figure 2 illustrates the four simulated environments: single-vehicle tracking (S1), crossing trajectories (S2), occlusion (S3), and shared-zone conflict awareness (S4).



(a) S1: single-car tracking.



(b) S2/S4: crossing and shared-zone awareness.



(c) S3: occlusion scenario.

Fig. 2: Simulated Gazebo scenarios for single-target tracking, multi-vehicle interaction, occlusion, and UAV-assisted shared-zone awareness.

TABLE I: Evaluated scenarios.

Scenario	Description	Main challenge
S1	Single car	Basic aerial tracking
S2	Crossing	Interaction and crossing trajectories
S3	Occlusion	Visual interruption and continuity
S4	Shared-zone approach	Conflict-awareness lead time

TABLE II: Evaluation validity checks.

Check	Color	YOLO	Fusion
Total runs	200	200	200
Runs per scenario	50	50	50
Metric files generated	200	200	200
YOLO loaded correctly	N/A	200	200
Ultralytics failures	N/A	0	0
Color proxy fallback	N/A	0	0

E. Metrics

The tracking methods were evaluated using the following metrics:

- **Detection coverage:** fraction of samples with a visual detection.
- **Track coverage:** fraction of samples with an active target track.
- **RMSE:** root mean squared position error.
- **MAE:** mean absolute position error.
- **Position standard deviation:** variability of the position error.
- **Conflict lead time:** time between first track availability and the instant at which the distance between the simulated UGV reference states first becomes smaller than 0.75 m.

Visual detections remain valid for up to 0.5 s, potentially increasing track coverage beyond detection coverage. While detection coverage measures visual availability, track coverage measures target-estimate continuity, the main objective of the proposed fusion method.

Localization error is evaluated against the idealized simulated UGV state, which is also available to the fusion method.

IV. Results

A. Evaluation Validity

All three methods completed the full evaluation matrix. Each method produced 200 valid runs, with 50 runs per scenario. The YOLO baseline and the proposed fusion method loaded the trained YOLO model correctly in all runs, with no fallback to the color proxy detector. Table II summarizes the evaluation validity.

B. Quantitative Results

Table III presents the full quantitative comparison. The color baseline provides a simple visual reference, but its localization error is high. The YOLO baseline substantially improves localization accuracy when the target is visible. The proposed UAV–UGV fusion method achieves the highest track coverage and the lowest RMSE among the evaluated methods.

Figure 3 summarizes the main quantitative trends obtained from the full evaluation matrix. The left plot compares localization error, while the right plot compares track coverage across the four scenarios. The results are consistent with Table III: the color baseline is able to detect the marker in some scenarios, but its localization error remains high; the YOLO baseline substantially improves localization accuracy when the target is visible; and the proposed UAV–UGV fusion method achieves the highest measured track coverage together with the lowest RMSE. In particular, the fusion method obtains the lowest RMSE and the highest track coverage in all scenarios, indicating that combining visual detections with shared UGV state improves tracking continuity and localization accuracy under the evaluated simulation conditions.

Figure 4 analyzes the same behavior from a temporal perspective by estimating the lost-track time from the measured track coverage. The visual-only baselines lose the target for longer periods as scenario complexity increases, especially in the crossing, occlusion, and shared-awareness scenarios. In contrast, the proposed fusion method maintains a substantially lower lost-track time because shared UGV state information preserves the target estimate during visual gaps. This result supports the main hypothesis of the paper: visual detection alone is useful when the target is visible, while heterogeneous

TABLE III: Main quantitative results across all scenarios. Values are reported as mean ± standard deviation over 50 runs per scenario.

Scenario	Method	Detection coverage	Track coverage	RMSE (m)	MAE (m)
S1 single car	Color baseline	$\mathbf{68.3 \pm 1.3}\%$	$68.2 \pm 1.3\%$	3.640 ± 0.002	3.640 ± 0.002
S1 single car	YOLO baseline	$64.9 \pm 13.8\%$	$63.1 \pm 16.2\%$	0.022 ± 0.017	0.017 ± 0.017
S1 single car	UAV–UGV fusion	$63.0 \pm 16.3\%$	$\mathbf{99.9 \pm 0.2}\%$	$\mathbf{0.014 \pm 0.006}$	$\mathbf{0.010 \pm 0.005}$
S2 crossing	Color baseline	$\mathbf{58.0 \pm 5.9}\%$	$59.9 \pm 6.0\%$	6.143 ± 0.189	5.405 ± 0.156
S2 crossing	YOLO baseline	$34.9 \pm 0.5\%$	$35.3 \pm 0.5\%$	2.531 ± 0.023	2.294 ± 0.023
S2 crossing	UAV–UGV fusion	$31.5 \pm 8.4\%$	$\mathbf{99.9 \pm 0.1}\%$	$\mathbf{0.905 \pm 0.232}$	$\mathbf{0.476 \pm 0.126}$
S3 occlusion	Color baseline	$26.7 \pm 26.9\%$	$27.8 \pm 28.1\%$	6.436 ± 0.008	5.786 ± 0.009
S3 occlusion	YOLO baseline	$\mathbf{35.6 \pm 0.1}\%$	$38.2 \pm 0.2\%$	2.600 ± 0.007	2.379 ± 0.010
S3 occlusion	UAV–UGV fusion	$32.3 \pm 6.9\%$	$\mathbf{99.9 \pm 0.1}\%$	$\mathbf{0.974 \pm 0.204}$	$\mathbf{0.542 \pm 0.116}$
S4 UAV-assisted awareness	Color baseline	$0.0 \pm 0.0\%$	$0.0 \pm 0.0\%$	N/A	N/A
S4 UAV-assisted awareness	YOLO baseline	$\mathbf{34.8 \pm 0.3}\%$	$34.8 \pm 0.3\%$	1.987 ± 0.010	1.952 ± 0.011
S4 UAV-assisted awareness	UAV–UGV fusion	$34.8 \pm 0.4\%$	$\mathbf{99.9 \pm 0.1}\%$	$\mathbf{0.760 \pm 0.006}$	$\mathbf{0.443 \pm 0.006}$

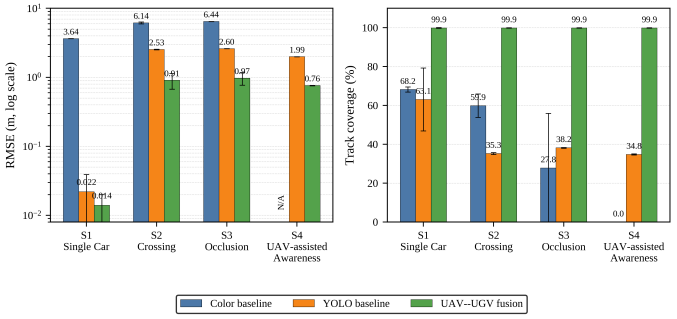


Fig. 3: Quantitative comparison across scenarios using the collected evaluation metrics. The proposed UAV–UGV fusion method reduces localization error and increases track coverage compared with the color and YOLO visual baselines.

TABLE IV: Improvement of the proposed method over the YOLO baseline.

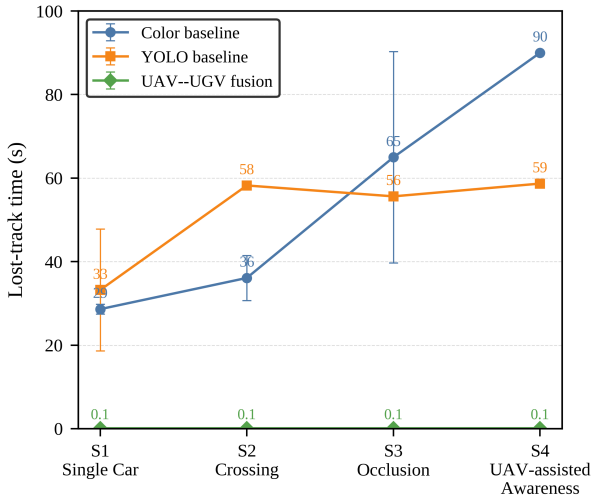
Scenario	Track gain	RMSE reduction	MAE reduction
S1	+36.7 pp	37.3%	39.3%
S2	+64.5 pp	64.2%	79.2%
S3	+61.7 pp	62.6%	77.2%
S4	+65.1 pp	61.8%	77.3%

UAV–UGV state sharing improves target-state continuity under intermittent observations.

C. Improvement Over the YOLO Baseline

Table IV reports the relative improvement of the proposed method over the YOLO visual baseline. Since both methods use the same trained YOLO detector, these differences isolate the effect of adding shared UGV state information to the tracking pipeline.

The largest improvement appears in track coverage. In S2, S3, and S4, the YOLO baseline tracks the target for only about one third of the trial because visual detections are available only during part of the trajectory. The proposed fusion method maintains nearly continuous tracking by using the shared UGV state whenever visual detections are unavailable. This also reduces localization error, particularly in the more challenging multi-vehicle and occlusion scenarios.



	RMSE	MAE	Detection	Track	Lost-track
S1 Single Car	Fusion	Fusion	Color	Fusion	Fusion
S2 Crossing	Fusion	Fusion	Color	Fusion	Fusion
S3 Occlusion	Fusion	Fusion	YOLO	Fusion	Fusion
S4 Awareness	Fusion	Fusion	Tie	Fusion	Fusion

Fusion best YOLO best Color best Tie

Fig. 4: Temporal robustness comparison based on lost-track time estimated from the collected track coverage. The proposed UAV–UGV fusion method maintains lower lost-track time across all scenarios.

D. Conflict Awareness

The interaction scenarios S2 and S4 also evaluate conflict-awareness lead time. In the present evaluation, this metric is defined as the interval between first track availability and the instant at which the distance between the simulated UGV reference states first becomes smaller than 0.75 m. No valid

TABLE V: Conflict-awareness lead time in interaction scenarios.

Scenario	Color	YOLO	Fusion
S2 crossing	N/A	N/A	$\mathbf{10.20 \pm 0.22}$ s
S4 awareness	N/A	N/A	$\mathbf{9.07 \pm 0.06}$ s

lead-time estimate was obtained for the visual baselines under the adopted metric, while the proposed fusion method maintained target-state availability before the vehicles entered the predefined conflict region. Table V summarizes these results.

These results indicate that the proposed method maintains target-state availability earlier in the interaction scenarios. By maintaining a continuous target estimate, the fusion approach provides tracking information several seconds before the vehicles enter the predefined conflict region.

E. Discussion

The color baseline provides a useful training-free reference, but its position estimates are inaccurate under perspective changes and multi-robot configurations. Although it reaches moderate detection coverage in S1 and S2, its RMSE remains much higher than the learned and fusion-based approaches. In S4, the color baseline produces no valid detections, indicating that simple marker segmentation is not sufficiently robust for the shared-awareness setting.

The YOLO baseline substantially improves localization accuracy when the target is visible. In S1, it achieves an RMSE of approximately 0.022 m, indicating low localization error under favorable visibility. However, the YOLO baseline remains limited by the UAV camera field of view. In S2, S3, and S4, track coverage remains close to 35–38%, reflecting the portions of the trajectory in which the target is actually observed by the UAV.

The proposed UAV–UGV fusion method mitigates this limitation by using shared UGV state information to maintain the target estimate during visual gaps. As a result, it achieves nearly 100% track coverage across all scenarios and reduces RMSE relative to the YOLO baseline in every case. The largest gains occur in the crossing, occlusion, and shared-awareness scenarios, where visual-only tracking is most affected by temporary loss of visibility.

The present evaluation assumes reliable access to the simulated UGV state and does not explicitly model localization noise, communication delay, packet loss, or asynchronous state updates. Therefore, the reported results characterize the proposed method under reliable state-sharing conditions.

V. Conclusion

This paper presented a heterogeneous UAV–UGV methodology for aerial tracking of mini autonomous cars. The system was implemented in ROS 2 and Gazebo Sim using a Tello-like UAV and multiple UGVs. Three tracking approaches were compared. These included a training-free color baseline, a YOLO visual baseline, and a UAV–UGV fusion method.

The experiments showed that visual detection alone is effective while the target remains visible. Track continuity decreased when the target left the camera field of view or became occluded. The proposed fusion method maintained target estimates during these visual gaps by combining visual detections with shared UGV state information. This strategy improved tracking continuity without modifying the visual detector.

The proposed method achieved higher track coverage than both visual baselines across all evaluated scenarios. The largest improvements were observed during crossing trajectories, occlusions, and shared-awareness scenarios. Under the evaluated simulation conditions, these results indicate that heterogeneous state sharing improves target-state continuity when visual observations are intermittent. The proposed methodology also provides a reproducible framework for evaluating heterogeneous UAV–UGV state sharing under controlled simulation conditions.

The YOLO baseline and fusion method use the same trained detector; the fusion method additionally incorporates shared UGV state information. Thus, the reported differences isolate the contribution of heterogeneous state sharing relative to the YOLO visual baseline. However, since the evaluation uses idealized simulated UGV state information as its reference, the reported localization gains do not establish robustness to localization uncertainty.

Future work will evaluate the proposed methodology on real robotic platforms. The influence of sensor noise and communication delays will also be investigated. The idealized simulated UGV state will be replaced with onboard localization estimates. Additional experiments will address multi-target identity management. Future work will also investigate UAV motion planning strategies that actively preserve target visibility.

References

[1] Y. Zhou, J. Zhao, N. Sun, Y. Wu, Q. Shi, G. Wang, and A. El Kamel, "Heterogeneous agents, unified missions: A survey and taxonomy on air–ground cooperative systems," *Robotics and Autonomous Systems*, vol. 203, p. 105514, 2026.

[2] I. D. Miller, F. Cladera, T. Smith, C. J. Taylor, and V. Kumar, "Stronger together: Air–ground robotic collaboration using semantics," *arXiv preprint arXiv:2206.14289*, 2022.

[3] P. Moraes, M. Rodriguez, K. S. Kappel, H. Sodre, S. Fernandez, I. Nunes, B. Guterres, and R. Grando, "Mini autonomous car driving based on 3d convolutional neural networks," in *2025 Brazilian Symposium on Robotics (SBR) and 2025 Workshop on Robotics in Education (WRE)*. IEEE, 2025, pp. 206–212.

[4] J. Redmon, S. Divvala, R. Girshick, and A. Farhadi, "You only look once: Unified, real-time object detection," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 2016, pp. 779–788.

[5] G. Jocher, A. Chaurasia, and J. Qiu, "Ultralytics yolov8," 2023. [Online]. Available: https://github.com/ultralytics/ultralytics

[6] P. Wu, Y. Li, and D. Xue, "Uav target tracking: A survey," *Artificial Intelligence Review*, vol. 58, no. 358, 2025.

[7] A. Barisic, M. Car, and S. Bogdan, "Vision-based system for a real-time detection and following of uav," *arXiv preprint arXiv:2205.00083*, 2022.

[8] M. Yaseen, "What is yolov8: An in-depth exploration of the internal features of the next-generation object detector," *arXiv preprint arXiv:2408.15857*, 2024.

[9] D. Reis, J. Kupec, J. Hong, and A. Daoudi, "Real-time flying object detection with yolov8," *arXiv preprint arXiv:2305.09972*, 2023.